Research Article

Fuzzy Logic-Based Queuing Models for Uncertain and Imprecise Queueing Environments

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Abstract

Classical queueing models typically assume precise arrival and service parameters, which may be unrealistic in systems characterized by uncertainty and imprecise information. In the present manuscript, a fuzzy logic-based queueing framework in which uncertain arrival and service rates are represented by fuzzy numbers and linguistic variables is developed. The proposed framework analyzes the effects of fuzzy parameters on key performance measures, including queue length, waiting time, system utilization, and server performance. Different fuzzy queueing configurations are investigated and compared with their corresponding classical models to evaluate the impact of parameter uncertainty on system behaviour. Numerical and computational analyses demonstrate how variations in fuzzy parameters and membership functions influence queueing performance. The results indicate that fuzzy queueing models provide a more flexible and realistic representation of queueing systems when precise parameter information is unavailable. Some suitable examples are also provided to validate the model.

Keywords: Fuzzy logic, fuzzy queuing model, uncertainty, imprecise data, waiting time, queue length, performance measures, arrival rate, service rate, queueing systems.

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1. Introduction

Queuing theory is an important branch of applied probability and operations research concerned with the mathematical analysis of waiting lines and congestion phenomena. Queueing systems arise naturally in a wide range of practical environments, including healthcare facilities, banking systems, transportation networks, telecommunication systems, manufacturing units, computer networks, and service organizations. The fundamental objective of queueing analysis is to determine and optimize important performance measures such as the expected number of customers in the queue, expected number of customers in the system, waiting time, system utilization, and probability of congestion. Classical queueing theory generally describes a system through probabilistic assumptions concerning customer arrivals and service processes. Kendall's notation provides a standard framework for characterizing queueing systems according to their arrival process, service-time distribution, and number of servers [1].

Traditional queueing models have been highly successful in situations where sufficient and reliable statistical information about arrival and service processes is available. However, many real-life queueing environments are characterized not only by randomness but also by vagueness, subjectivity, incomplete information, and imprecision. For example, the arrival rate at a hospital emergency department may be described by experts as "high" or "moderate" rather than by a precisely known numerical value. Similarly, service times in banks, hospitals, call centres, and repair facilities may fluctuate because of human behaviour and operational conditions. Representing such information by a single crisp parameter can therefore lead to unrealistic assumptions and may reduce the practical applicability of conventional queueing models.

Fuzzy set theory, introduced by Zadeh [2], provides a mathematical framework for representing information whose boundaries are not precisely defined. Unlike classical set theory, in which an element either belongs or does not belong to a set, fuzzy set theory assigns each element a membership grade in the interval [0,1]. This feature makes fuzzy methodology particularly suitable for modelling linguistic and imprecise information such as "low arrival rate," "moderate service time," "approximately five customers per minute," or "very long waiting time." Fuzzy logic consequently provides an alternative mechanism for handling uncertainty that is fundamentally different from probability-based randomness.

The integration of fuzzy set theory with queueing theory has led to the development of fuzzy queueing models. Early investigations demonstrated that uncertain inter-arrival and service information can be represented through fuzzy quantities rather than crisp parameters. This approach allows the performance measures of a queue to retain the uncertainty present in the input data. Li and Lee [3] investigated fuzzy queueing systems and demonstrated the use of fuzzy approaches for analysing queueing performance. Negi and Lee [4] combined fuzzy-set methodology with conventional queueing approaches and investigated both analytical and simulation-based procedures for fuzzy queues.

An important development in fuzzy queueing theory is the use of α-cuts for transforming a fuzzy queueing problem into a family of corresponding crisp queueing problems. Kao, Li, and Chen [5] proposed a parametric programming approach for constructing membership functions of queueing performance measures when inter-arrival and service times are fuzzy. Their methodology considered several fuzzy queueing structures, including M/F/1, F/M/1, F/F/1, and FM/FM/1 models.

The importance of fuzzy queueing models is particularly evident when performance measures themselves are uncertain. In a conventional queue, expected waiting time or queue length is generally represented by a crisp value obtained from known model parameters. In contrast, when arrival and service characteristics are fuzzy, the resulting waiting time, queue length, system length, and utilization may also be fuzzy quantities. Thus, the output of the model can provide a range of possible performance levels together with their corresponding membership grades rather than a single potentially misleading value [4,5].

Several studies have further demonstrated the applicability of fuzzy queueing theory to practical systems. Fuzzy queueing methods have been applied to computer-system reliability and failure analysis, where uncertain failure-arrival processes and service characteristics can be incorporated into fuzzy queueing frameworks [6]. Fuzzy queueing has also been used in construction and simulation environments where subjective uncertainty in activity durations produces uncertainty in inter-arrival and service times [7].

Despite these developments, there remains a need for systematic frameworks that examine the influence of fuzzy parameters across different queueing configurations and explicitly compare fuzzy and conventional queueing models. In particular, the sensitivity of waiting time, queue length, utilization, and other performance measures to changes in the membership functions of arrival and service parameters deserves further investigation.

Motivated by these considerations, the present study develops a fuzzy logic-based framework for analysing queueing behaviour under uncertain and imprecise operating conditions. The study considers fuzzy representations of important queueing parameters and investigates their influence on key performance measures. Different queueing configurations are considered to demonstrate the flexibility of the proposed approach. The resulting fuzzy performance measures are then compared with their classical crisp counterparts to assess the extent to which fuzzy modelling can provide a more informative representation of queueing behaviour.

The main objectives of this study are:

To apply fuzzy logic to queueing models for handling uncertain and imprecise input information.

To formulate fuzzy-based models for different types of queueing systems.

To investigate the influence of fuzzy parameters on important performance measures, particularly waiting time and queue length.

To compare fuzzy queueing models with conventional queueing models in order to evaluate their relative accuracy, flexibility, and efficiency in representing uncertain queueing environments.

The proposed framework is expected to provide a flexible mathematical methodology for situations in which precise probabilistic information is unavailable or insufficient. By retaining the uncertainty of the input parameters in the resulting performance measures, fuzzy queueing models can support more realistic analysis and decision-making in practical service and congestion systems.

2. Preliminaries

In this section, the basic concepts of queueing theory and fuzzy set theory required for the development of the proposed fuzzy queueing framework are introduced.

2.1. Basic Queueing System

A queueing system consists fundamentally of a population of customers requiring service, an arrival mechanism, one or more service facilities, a waiting line, and a service discipline. Customers arrive according to an arrival process and either receive service immediately or wait in a queue when all servers are occupied.

A conventional queueing system can be characterized by:

λ: mean arrival rate; μ: mean service rate of an individual server; c: number of servers; Lq: expected number of customers in the queue; Ls: expected number of customers in the system; Wq: expected waiting time in the queue; Ws: expected time spent in the system; ρ: utilization factor.

For a single-server system, $\rho = \lambda/\mu$, and the usual stability requirement is $\rho < 1$. For a c-server system, $\rho = \lambda/(c\mu)$, provided that all servers have the same service rate.

Kendall's notation is commonly used to represent queueing models. The basic form is A/S/c, where A represents the inter-arrival-time distribution, S represents the service-time distribution, and c denotes the number of servers [1]. For example, M/M/1 denotes a single-server queue with exponentially distributed inter-arrival and service times, whereas M/M/c represents a multi-server system with Poisson arrivals and exponential service times.

2.2. Queueing Performance Measures

The major performance measures considered in this study are the average queue length and average waiting time. Let Lq denote the expected number of customers waiting in the queue and Wq denote the expected waiting time before receiving service. Similarly, Ls denotes the expected number of customers in the complete system and Ws denotes the expected time spent in the system.

Little's law provides the fundamental relationship $L = \lambda W$. The corresponding relationship for the queue is $L_q = \lambda W_q$. These relationships establish a direct connection between queue length and waiting time and can subsequently be extended to fuzzy environments.

2.3. Fuzzy Set

Let X be a non-empty universe of discourse. A fuzzy set A~ on X is defined by a membership function $\mu A\sim: X \rightarrow [0,1]$, where $\mu A\sim(x)$ represents the degree to which x belongs to A~ [2]. Thus, $A\sim = \{(x, \mu A\sim(x)): x \in X\}$. The values 0 and 1 indicate no membership and complete membership, respectively.
In queueing applications, linguistic information such as "low arrival rate," "approximately five customers per minute," or "high service rate" can be represented by fuzzy sets.

2.4. Fuzzy Number

A fuzzy set A~ defined on R is called a fuzzy number if it is normal, convex, upper semicontinuous, and has bounded support. Fuzzy numbers are particularly useful in queueing applications because uncertain arrival and service rates can be represented directly as fuzzy quantities.

2.5. Triangular Fuzzy Number

A triangular fuzzy number is commonly represented as $A\sim = (a_1, a_2, a_3)$, where $a_1 \le a_2 \le a_3$. Its membership function is

$\mu A\sim(x) = 0, \quad x < a_1,$
$\quad (x - a_1)/(a_2 - a_1), \quad a_1 \le x \le a_2,$
$\quad (a_3 - x)/(a_3 - a_2), \quad a_2 \le x \le a_3,$
$\quad 0, \quad x > a_3.$

Here, a_1 and a_3 represent lower and upper bounds, while a2 represents the most plausible value. For example, an uncertain arrival rate may be represented by $\lambda\sim = (4,5,7)$.

2.6. Trapezoidal Fuzzy Number

A trapezoidal fuzzy number is represented as $A\sim = (a_1, a_2, a_3, a_4)$, where $a_1 \le a_2 \le a_3 \le a_4$. Its membership function is

$$\mu\sim A(x) = \begin{cases} 0 & x < a_1 \\ (x - a_1)/(a_2 - a_1) & a_1 \le x \le a_2 \\ 1 & a_2 \le x \le a_3 \\ (a_4 - x)/(a_4 - a_3) & a_3 \le x \le a_4 \\ 0 & x > a_4 \end{cases}$$

Trapezoidal fuzzy numbers are useful when a range of values is regarded as equally plausible rather than when there is a single most plausible value.

2.7. α-Cut of a Fuzzy Number

For a fuzzy set A~ and $\alpha \in [0,1]$, the α-cut is defined by
$[A\sim]\alpha = \{x \in X : \mu A\sim(x) \ge \alpha\}$.
For a fuzzy number, the α-cut is a closed interval $[A_L(\alpha), A_U(\alpha)]$. The α-cut converts a fuzzy parameter into an interval of possible crisp values. Consequently, a fuzzy queueing problem can be analysed as a family of conventional queueing problems corresponding to different values of α [5].
If the fuzzy arrival rate is λ~, then $[\lambda\sim]\alpha = [\lambda_L(\alpha), \lambda_U(\alpha)]$.
Similarly, $[\mu\sim]\alpha = [\mu_L(\alpha), \mu_U(\alpha)]$.

2.8. Fuzzy Arithmetic

Let A~ and B~ be fuzzy numbers. Their arithmetic operations can be defined using the extension principle. For practical computation, α-cut based interval arithmetic is frequently employed. For example,
$[A\sim + B\sim]\alpha = [A_L(\alpha) + B_L(\alpha), A_U(\alpha) + B_U(\alpha)]$.
For positive fuzzy numbers,
$[A\sim B\sim]\alpha = [A_L(\alpha)B_L(\alpha), A_U(\alpha)B_U(\alpha)]$.
The extension principle provides a systematic mechanism for propagating fuzzy input information through mathematical functions [2].

2.9. Fuzzy Arrival and Service Parameters

In a conventional queue, λ and μ are usually assumed to be crisp values. In the proposed fuzzy framework, these parameters can be represented by fuzzy numbers λ~ and μ~. The traffic intensity for a single-server system can therefore be expressed as
$\rho\sim = \lambda\sim/\mu\sim$.

For c identical servers,
$\rho\sim = \lambda\sim/(c\mu\sim)$.
The resulting utilization is a fuzzy quantity. System stability can consequently be examined at different α levels using interval representations.

2.10. Fuzzy Queueing Model

A fuzzy queueing model is obtained when one or more parameters of a conventional queueing model are represented by fuzzy quantities. For example, a conventional M/M/1 system can be extended by replacing λ and μ with λ~ and μ~. The performance measures consequently become fuzzy quantities:
$L_q \rightarrow L_{\sim q}$, $W_q \rightarrow W_{\sim q}$, $L_s \rightarrow L_{\sim s}$, $W_s \rightarrow W_{\sim s}$.
The membership functions of these performance measures describe the degree of possibility associated with different values of waiting time or queue length [5,7].

2.11. Fuzzy Performance Measures

The primary performance measures considered in the proposed study are the average fuzzy queue length L~q, average fuzzy waiting time W~q, average fuzzy system length L~s, average fuzzy system time W~s, and fuzzy utilization ρ~. The study investigates how variations in λ~ and μ~ influence these measures.

2.12. Relationship Between Fuzzy Inputs and Queueing Outputs

Let a conventional queueing performance measure be represented by Y = f(λ, μ, c). When λ and/or μ are fuzzy, the corresponding fuzzy performance measure can be expressed as Y~ = f(λ~, μ~, c). Using the extension principle,
$\mu Y\sim(y) = \sup \min\{ \mu\lambda\sim(x_1), \mu\mu\sim(x_2) \}$,
subject to $y = f(x_1,x_2)$.
In practice, α-cut and interval-based approaches provide an effective computational alternative. Kao et al. [5] demonstrated how fuzzy queues can be transformed into families of crisp queueing models at different possibility levels.

2.13. Fuzzy Versus Conventional Queueing Models

The fundamental difference between conventional and fuzzy queueing models lies in the representation of uncertain information. In a conventional model, λ = λ0 and μ = μ0, whereas in a fuzzy model λ = λ~ and μ = μ~. Consequently, the conventional model produces a single value for each performance measure, while the fuzzy model can produce a membership function or interval-valued performance measures.
This distinction is particularly useful when information is based on expert judgement or linguistic descriptions rather than sufficiently large historical datasets. Fuzzy queueing approaches can preserve the vagueness of input information and provide more informative performance descriptions for practical decision-making [3–7]. The proposed study builds upon these foundations by systematically examining different fuzzy queueing configurations, analysing the sensitivity of waiting time and queue length to fuzzy input parameters, and comparing the resulting fuzzy performance measures with those obtained from corresponding conventional queueing models.
Classical queueing models normally require crisp estimates of arrival and service rates. In many practical systems, however, the available information is imprecise: managers may describe demand as low, moderate, or high, while service capacity may be approximately known rather than exactly measured. Existing fuzzy queueing approaches address this problem by replacing crisp parameters with fuzzy quantities and deriving fuzzy performance measures. Important foundations include the analysis of fuzzy queues by Li and Lee, simulation and analysis by Negi and Lee, and the parametric programming approach of Kao, Li, and Chen [8-10].
The present paper proposes an Adaptive α-Cut Fuzzy Queueing (AFQ) model. The central idea is to combine fuzzy arrival and service rates with an α-cut interval analysis and an adaptive aggregation mechanism. Instead of treating every α-level identically, the proposed framework assigns greater analytical importance to α-levels where the input information is more reliable or where the resulting queueing performance is more stable. This produces a decision-oriented fuzzy description of waiting time and queue length while retaining the uncertainty of the input data.
The model should be regarded as a proposed framework requiring mathematical and computational validation rather than as an established theorem. Its novelty can subsequently be strengthened by proving properties, conducting sensitivity experiments, and benchmarking it on real or simulated data.
Proposed Adaptive α-Cut Fuzzy Queueing Model
Consider a single-server queue with uncertain arrival and service rates. Let the arrival rate and service rate be represented by triangular fuzzy numbers
$\lambda\sim = (\lambda_1, \lambda_2, \lambda_3)$, $\mu\sim = (\mu_1, \mu_2, \mu_3)$,
where $\lambda_1 \le \lambda_2 \le \lambda_3$ and $\mu_1 \le \mu_2 \le \mu_3$. At each $\alpha \in [0,1]$, define the corresponding α-cut intervals as
$[\lambda\sim]\alpha = [\lambda_L(\alpha), \lambda_U(\alpha)]$, $[\mu\sim]\alpha = [\mu_L(\alpha), \mu_U(\alpha)]$.
For a triangular fuzzy number,
$\lambda_L(\alpha) = \lambda_1 + \alpha(\lambda_2 - \lambda_1)$,
$\lambda_U(\alpha) = \lambda_3 - \alpha(\lambda_3 - \lambda_2)$,
and similarly for μ~.

The proposed model evaluates the conventional M/M/1 performance equations over the α-cut intervals. For each α, the traffic intensity interval is

$[\rho\sim]\alpha = [\lambda_L(\alpha)/\mu_U(\alpha), \lambda_U(\alpha)/\mu_L(\alpha)]$,

provided the denominator is positive. The stability condition is therefore checked conservatively through

$\lambda_U(\alpha) < \mu_L(\alpha)$.

When this condition holds, the queue-length and waiting-time bounds can be generated. For the M/M/1 structure,

$Lq = \lambda^2 / [\mu(\mu - \lambda)]$,

$Wq = \lambda / [\mu(\mu - \lambda)]$,

$L = \lambda / (\mu - \lambda)$,

and

$W = 1 / (\mu - \lambda)$.

The proposed fuzzy model obtains lower and upper performance values from the corresponding α-cut intervals. For example,

$[L_q]\alpha = [L_{q,L}(\alpha), L_{q,U}(\alpha)]$

and

$[W_q]\alpha = [W_{q,L}(\alpha), W_{q,U}(\alpha)]$.

The exact endpoint calculation can be implemented using interval arithmetic or validated optimization at each α-level, which is preferable when monotonicity cannot be guaranteed.

3. Adaptive α-Level Aggregation

A distinctive component of the proposed AFQ model is the adaptive weighting of α-levels. Let w(α) be a non-negative weight satisfying

$\int_0^1 w(\alpha)d\alpha = 1$.

Instead of imposing a uniform weighting, w(α) can be selected from the reliability of the input information. One possible choice is

$w(\alpha) = c \exp(-\kappa(1-\alpha))$,

where $\kappa \geq 0$ controls the emphasis placed on high-membership information and c is the normalization constant.

For a fuzzy performance measure Y~, an adaptive representative value can then be defined as

$Y_AFQ = \int_0^1 w(\alpha)\, Y_m(\alpha)d\alpha$,

where $Y_m(\alpha)$ may be the midpoint of the α-cut,

$Y_m(\alpha) = [Y_L(\alpha)+Y_U(\alpha)]/2$.

The corresponding uncertainty width can be summarized by

$D_Y = \int_0^1 w(\alpha)[Y_U(\alpha)-Y_L(\alpha)]d\alpha$.

Thus, the proposed model provides two decision-oriented outputs: an adaptive central performance estimate and an aggregate uncertainty measure. This allows two systems with similar average waiting times but very different uncertainty widths to be distinguished.

4. Conceptual Diagram of the Proposed Model

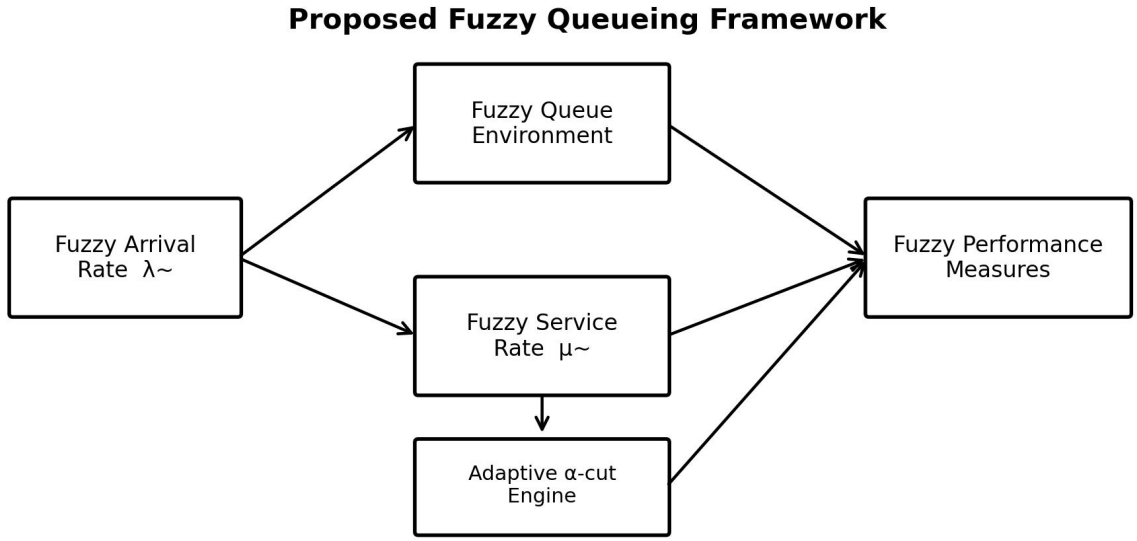


Figure 1. Proposed Adaptive α-Cut Fuzzy Queueing framework.

5. Computational Algorithm

The proposed model can be implemented using the following procedure:
Step 1. Specify the queueing structure, number of servers, and crisp queueing equations.
Step 2. Represent uncertain arrival and service parameters by suitable fuzzy numbers.
Step 3. Generate α-cuts for α = 0, 0.1, 0.2, …, 1 (or a finer grid).
Step 4. Check stability at every α-level using the lower service-rate and upper arrival-rate bounds.
Step 5. Compute lower and upper queueing performance measures.
Step 6. Calculate the midpoint and uncertainty width at every α-level.
Step 7. Apply the adaptive α-level weighting function.
Step 8. Obtain the adaptive representative performance and aggregate uncertainty.
Step 9. Compare the fuzzy results with the corresponding crisp M/M/1 results.
Step 10. Perform sensitivity analysis by changing the spreads of λ~ and μ~ and the adaptive parameter κ.

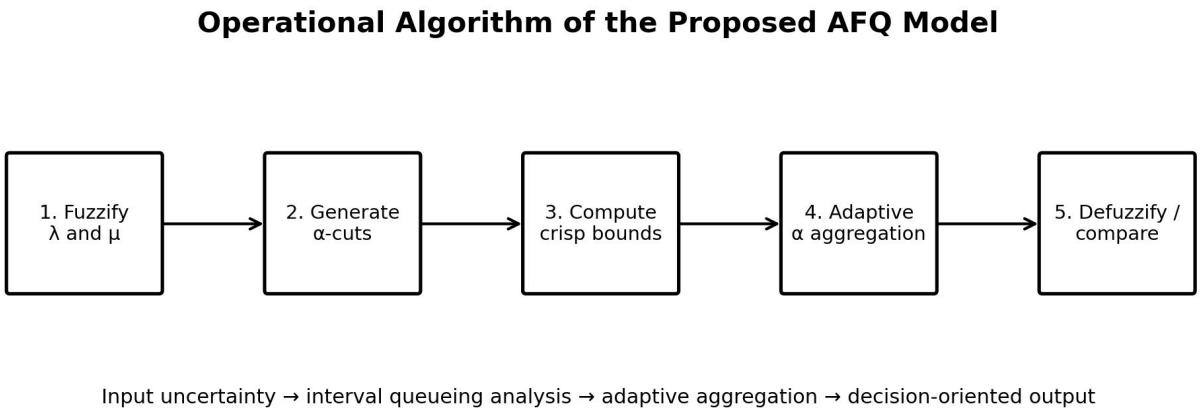


Figure 2. Computational workflow for the proposed AFQ model.

6. Illustrative Examples

For an illustrative study, consider an uncertain single-server service system with
λ~ = (4, 5, 7) customers per unit time,
and
μ~ = (8, 10, 13) customers per unit time.
The crisp benchmark may use λ = 5 and μ = 10, corresponding to the modal values. The conventional M/M/1 model then gives a traffic intensity of ρ = 0.50. The fuzzy model, however, retains the possible variation in arrival and service rates and therefore produces intervals for ρ, Lq, Wq, L, and W at every α-level.
For publication, the numerical experiment should be expanded to include several values of α and at least three uncertainty scenarios: low uncertainty, moderate uncertainty, and high uncertainty. This will allow the sensitivity of waiting time and queue length to the spreads of the fuzzy parameters to be demonstrated quantitatively.

6.1. α-Cuts of Fuzzy Parameters

For λ~ = (4,5,7), the α-cut is
$[\tilde{\lambda}]_\alpha = [\lambda_L(\alpha), \lambda_U(\alpha)]$,
where
$\lambda_L(\alpha) = 4 + \alpha(5 - 4) = 4 + \alpha$,
and
$\lambda_U(\alpha) = 7 - \alpha(7 - 5) = 7 - 2\alpha$.
Therefore,
$[\tilde{\lambda}]_\alpha = [4 + \alpha, 7 - 2\alpha]$.
Similarly, for μ~ = (8,10,13),
$\mu_L(\alpha) = 8 + 2\alpha$,
$\mu_U(\alpha) = 13 - 3\alpha$,
and hence
$[\tilde{\mu}]_\alpha = [8 + 2\alpha, 13 - 3\alpha]$.

6.2. Fuzzy Traffic Intensity

For an M/M/1 queue,
$\rho = \lambda/\mu$.

Since λ and μ are fuzzy,
$\tilde{\rho} = \tilde{\lambda}/\tilde{\mu}$.
Because the parameters are positive,
$[\tilde{\rho}]_\alpha = [\ \lambda_L(\alpha)/\mu_U(\alpha),\ \lambda_U(\alpha)/\mu_L(\alpha)\ ]$
$= [\ (4+\alpha)/(13-3\alpha),\ (7-2\alpha)/(8+2\alpha)\ ].$
At $\alpha = 0.5$,
$\lambda_L = 4.5$, $\lambda_U = 6$, $\mu_L = 9$, and $\mu_U = 11.5$.
Thus,
$\rho_L = 4.5/11.5 = 0.3913$,
$\rho_U = 6/9 = 0.6667$.
Hence,
$[\tilde{\rho}]_{0.5} = [0.3913, 0.6667]$.
The utilization remains below one at this α-level, indicating stability.

6.3. Fuzzy Queue Length

For an M/M/1 queue,
$L_q = \lambda^2 / [\mu(\mu-\lambda)]$.
Thus,
$\tilde{L}_q = \tilde{\lambda}^2 / [\tilde{\mu}(\tilde{\mu}-\tilde{\lambda})]$.
Since L_q increases with λ and decreases with μ,
$[\tilde{L}_q]_\alpha = [\lambda_L^2 / \{\mu_U(\mu_U-\lambda_L)\}, \lambda_U^2 / \{\mu_L(\mu_L-\lambda_U)\}]$.
Therefore,
$[\tilde{L}_q]_\alpha = [(4+\alpha)^2 / \{(13-3\alpha)(9-4\alpha)\}, (7-2\alpha)^2 / \{(8+2\alpha)(1+4\alpha)\}]$.
At $\alpha = 0.5$,
$L_{q,L} = 4.5^2/[11.5(11.5-4.5)] = 20.25/80.5 = 0.2516$,
and
$L_{q,U} = 6^2/[9(9-6)] = 36/27 = 1.3333$.
Therefore,
$[\tilde{L}_q]_{0.5} = [0.2516, 1.3333]$.

6.4. Fuzzy Waiting Time

For an M/M/1 queue,
$W_q = \lambda / [\mu(\mu-\lambda)]$.
Therefore,
$[\tilde{W}_q]_\alpha = [\lambda_L / \{\mu_U(\mu_U-\lambda_L)\}, \lambda_U / \{\mu_L(\mu_L-\lambda_U)\}]$.
Thus,
$[\tilde{W}_q]_\alpha = [\ (4+\alpha)/\{(13-3\alpha)(9-4\alpha)\}, (7-2\alpha)/\{(8+2\alpha)(1+4\alpha)\}]$.
At $\alpha = 0.5$,
$W_{q,L} = 4.5/[11.5(11.5-4.5)] = 4.5/80.5 = 0.0559$,
and
$W_{q,U} = 6/[9(9-6)] = 6/27 = 0.2222$.
Hence,
$[\tilde{W}_q]_{0.5} = [0.0559, 0.2222]$.

6.5. Proposed Uncertainty Indices

A distinctive component of the proposed model is the explicit quantification of uncertainty.
Define the Queue-Length Uncertainty Index by
$UI_L(\alpha) = L_{q,U}(\alpha) - L_{q,L}(\alpha)$,
and the Waiting-Time Uncertainty Index by
$UI_W(\alpha) = W_{q,U}(\alpha) - W_{q,L}(\alpha)$.
At $\alpha = 0.5$,
$UI_L(0.5) = 1.3333 - 0.2516 = 1.0817$,
and
$UI_W(0.5) = 0.2222 - 0.0559 = 0.1663$.
Therefore,
$UI_L(0.5) = 1.0817$,
$UI_W(0.5) = 0.1663$.

6.6. Adaptive Reliability Weight

Introduce the adaptive reliability-weighting function
$w(\alpha) = e^{\kappa\alpha} / \int_0^1 e^{\kappa u}\, du$,
where $\kappa \geq 0$ controls the emphasis placed on higher membership levels.
For $\kappa = 2$,
$\int_0^1 e^{2u}\, du = (e^2-1)/2$,

and therefore
$w(\alpha) = 2e^{2\alpha}/(e^2-1)$.
The weighting function gives greater importance to higher α-level information.

6.7. Adaptive Queue-Length and Waiting-Time Measures

Define the midpoint of the fuzzy queue length as
$L_m(\alpha) = [L_{q,L}(\alpha) + L_{q,U}(\alpha)]/2$.
The proposed Adaptive Fuzzy Queue Length is

$L_{AF} = \int_0^1 w(\alpha)L_m(\alpha)d\alpha$.
Similarly,
$W_{AF} = \int_0^1 w(\alpha)W_m(\alpha)d\alpha$,
where
$W_m(\alpha) = [W_{q,L}(\alpha) + W_{q,U}(\alpha)]/2$.
Using numerical integration over α = 0, 0.1, …, 1 for κ = 2 gives approximately
$L_{AF} \approx 0.7865$,
and
$W_{AF} \approx 0.1383$.

6.8. Aggregate Uncertainty

The aggregate queue-length uncertainty is defined by

$AUI_L = \int_0^1 w(\alpha)[L_{q,U}(\alpha) - L_{q,L}(\alpha)]d\alpha$.
For the numerical example,
$AUI_L \approx 0.9091$.
Similarly,

$AUI_W = \int_0^1 w(\alpha)[W_{q,U}(\alpha) - W_{q,L}(\alpha)]d\alpha$,
giving approximately
$AUI_W \approx 0.1363$.
The proposed framework therefore reports both performance and uncertainty.

6.9. α-Level Numerical Results

α	λ_L	λ_U	μ_L	μ_U	$L_{q,L}$	$L_{q,U}$	$W_{q,L}$	$W_{q,U}$
0.0	4.0	7.0	8.0	13.0	0.1368	6.1250	0.0342	0.8750
0.1	4.1	6.8	8.2	12.7	0.1539	4.0279	0.0375	0.5923
0.2	4.2	6.6	8.4	12.4	0.1735	2.8810	0.0413	0.4365
0.3	4.3	6.4	8.6	12.1	0.1959	2.1649	0.0456	0.3383
0.4	4.4	6.2	8.8	11.8	0.2217	1.6801	0.0504	0.2710
0.5	4.5	6.0	9.0	11.5	0.2516	1.3333	0.0559	0.2222
0.6	4.6	5.8	9.2	11.2	0.2863	1.0754	0.0622	0.1854
0.7	4.7	5.6	9.4	10.9	0.3269	0.8779	0.0695	0.1568
0.8	4.8	5.4	9.6	10.6	0.3748	0.7232	0.0781	0.1339
0.9	4.9	5.2	9.8	10.3	0.4317	0.5998	0.0881	0.1154
1.0	5.0	5.0	10.0	10.0	0.5000	0.5000	0.1000	0.1000

6.10. Important Numerical Observation

At α = 0,
$L_q \in [0.1368, 6.1250]$,
whereas at α = 1,
$L_q = 0.5000$.
Thus, as the membership level increases, the uncertainty interval contracts from
[0.1368, 6.1250] → [0.5000, 0.5000].
This provides a natural basis for studying the sensitivity of queue length and waiting time to the degree of membership and uncertainty in the input parameters.

6.11. Comparison with the Classical M/M/1 Model

Using the modal values λ = 5 and μ = 10, the classical M/M/1 model gives
$\rho = 5/10 = 0.5$,
$L_q = 5^2/[10(10-5)] = 0.5$,
and
$W_q = 5/[10(10-5)] = 0.1$.
The comparison is therefore:

Model	Queue length	Waiting time
Classical M/M/1	0.5000	0.1000
Fuzzy model at α=0.5	[0.2516, 1.3333]	[0.0559, 0.2222]

Proposed adaptive model	0.7865	0.1383

6.12. Relative Uncertainty Ratio

To compare uncertainty relative to representative performance, define
$RUI_L = AUI_L/LAF$,
and
$RUI_W = AUI_W/WAF$.
For the present example,
$RUI_L = 0.9091/0.7865 \approx 1.156$,
and
$RUI_W = 0.1363/0.1383 \approx 0.986$.
These indicators show that uncertainty is substantial relative to the representative queueing performance. The Relative Uncertainty Ratio can therefore be used as an additional decision-support indicator when comparing alternative queueing systems.

7. Comparison with Previous Fuzzy Queueing Models

Model	Main idea	Fuzzy inputs	Performance output	Limitation / distinction
Li & Lee [8]	Analytical analysis of fuzzy queues	Fuzzy queueing quantities	Fuzzy queue measures	Foundational framework; less emphasis on adaptive α-level decision weighting.
Negi & Lee [9]	Analysis and simulation of fuzzy queues	Fuzzy arrival/service information	Fuzzy performance through analysis/simulation	Simulation-oriented; adaptive uncertainty aggregation is not the central feature.
Kao, Li & Chen [10]	Parametric programming	Fuzzy inter-arrival/service times	Membership functions of queue measures	Strong α-cut/parametric foundation; representative decision score is not the main focus.
Proposed AFQ model	Adaptive α-cut fuzzy queueing	Fuzzy λ and μ	α-cut intervals + adaptive representative value + uncertainty width	Adds reliability-sensitive α weighting and an explicit uncertainty-width measure for comparison/decision support.

8. Conclusion

The proposed model handles imprecise arrival and service information without forcing a single crisp estimate. It produces interval-valued waiting time and queue-length information at multiple α-levels. Further it provides an adaptive representative performance value rather than relying only on a single defuzzification rule. Also, it reports an explicit uncertainty-width measure, making model uncertainty visible to decision-makers. This model can be applied to M/M/1 and extended to M/M/c, M/G/1, finite-capacity, priority, and other queueing structures. Also it provides a natural framework for sensitivity analysis by changing fuzzy spreads and the adaptive parameter κ.

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